Sliding-Window Complex Rules and Simple Functional Profiles

Companion note for `ar_process_sliding_window_embedding.ipynb`

1 Purpose

The notebook constructs a sequence of simplicial complexes

$$L_1, L_2, \ldots, L_{k_{\max}}$$

from a simulated autoregressive time series. For each embedding scale k, the notebook first builds a graph using one of the selected construction rules, then takes the flag complex of that graph. The plotted profile is a simple functional

$$k \longmapsto F(L_k), \qquad k = 1, \ldots, k_{\max}.$$

The following basic functionals are kept in the current notebook:

$$f_0, \quad f_1, \quad f_2, \quad \beta_0, \quad \beta_1, \quad \mathrm{PC}\,.$$

These are the entries of the face vector, the first two Betti numbers, and the point complexity. The notebook lets the user choose two such functionals F_1, F_2 and two scalar weights a, b, and plots

$$aF_1(L_k) + bF_2(L_k).$$

It also builds a scale-similarity matrix from the same linear combination:

$$S_{k\ell} = aF_1(L_k \oplus L_\ell) + bF_2(L_k \oplus L_\ell),$$

where $\oplus$ is either union or intersection, and plots the principal eigenvector of this matrix against k.

2 Time-Series and Lifting Notation

The simulated autoregressive process has order p:

$$x_t = c + \phi_1 x_{t-1} + \cdots + \phi_p x_{t-p} + \varepsilon_t.$$

In the interactive notebook, the user selects p, not the coefficient vector. The coefficients are drawn reproducibly from the seed in partial-autocorrelation coordinates. Since every partial autocorrelation lies in $(-1, 1)$, the resulting AR polynomial is stationary.

For a univariate series $x_0, \ldots, x_{n-1}$, the forward sliding-window embedding of dimension k is

$$Y_t^{(k)} = (x_t, x_{t+1}, \ldots, x_{t+k-1})^\top \in \mathbb{R}^k.$$

The series is split chronologically into training and test segments. For each $k \le k_{\max}$, the training windows of length $k_{\max}$ are split as

$$W_t = (x_t, \ldots, x_{t+k_{\max}-1})^\top = \begin{pmatrix} Y_t^{(k)} \\ Z_t^{(k)} \end{pmatrix},$$

where

$$Z_t^{(k)} = (x_{t+k}, \ldots, x_{t+k_{\max}-1})^\top.$$

The completion matrix $\widehat{A}_k$ is fitted by least squares:

$$\widehat{A}_k \in \arg\min_A \sum_t \left\| Z_t^{(k)} - A Y_t^{(k)} \right\|_2^2.$$

For $k = k_{\max}$, no tail is missing, and $\widehat{A}_{k_{\max}}$ is empty.

On the test segment, each k-window is lifted to the common ambient space $\mathbb{R}^{k_{\max}}$:

$$p_t^{(k)} = \begin{pmatrix} Y_t^{(k)} \\ \widehat{A}_k Y_t^{(k)} \end{pmatrix}.$$

The corresponding prediction-error vector is

$$e_t^{(k)} = \widehat{A}_k Y_t^{(k)} - Z_t^{(k)}.$$

3 Common Graph and Flag-Complex Setup

Fix k. Let m be the number of complete test windows of length $k_{\max}$. Each time index $t = 0, \ldots, m-1$ is a vertex:

$$V_k = \{0, 1, \ldots, m-1\}.$$

Before applying a construction rule, the notebook restricts attention to temporal candidate edges:

$$E_k^{\text{temp}} = \big\{\{s,t\} : 0 \le s < t \le m-1,\ 1 \le |s-t| \le k-1\big\}.$$

For $k = 1$, this set is empty. For larger k, a vertex can connect only to nearby window-start times.

Each construction rule produces a retained edge set

$$E_k \subseteq E_k^{\text{temp}}.$$

The graph is $H_k = (V_k, E_k)$. The simplicial complex is the flag complex

$$L_k = \text{Flag}(H_k).$$

Thus a finite vertex set $\sigma \subseteq V_k$ is a simplex of L_k exactly when every pair of vertices in σ is an edge of H_k. In the notebook, GUDHI expands the graph through dimension two.

4 Complex Construction Rules

The notebook now uses manually chosen thresholds. Let $\tau_e \ge 0$ be the error threshold, let $\tau_d \ge 0$ be the lifted-cloud distance threshold, and let $\tau_c \ge 0$ be the covariance threshold.

A general way to describe the construction is to put nonnegative weights on the temporal candidate graph. For a weight function

$$w_k : E_k^{\text{temp}} \to \mathbb{R}_{\ge 0},$$

retain the thresholded edge set

$$F_k^\tau = \{e \in E_k^{\text{temp}} : w_k(e) \le \tau\}.$$

This gives a subgraph

$$H_k = (V_k, F_k^\tau)$$

and therefore the data-dependent subcomplex

$$L_k = \mathrm{Flag}(H_k).$$

If one wants the sequence $L_1, \ldots, L_{k_{\max}}$ to be a filtration under the common temporal edge sets, the edge weights should satisfy

$$w_{k+1}(e) \le w_k(e), \qquad e \in E_k^{\mathrm{temp}}, \quad k = 1, \ldots, k_{\max} - 1.$$

The notebook reports the complexes scale by scale; this monotonicity condition is the mathematical condition to check when the selected rule is intended to define a true filtration.

4.1 Both Errors Small

Let

$$r_t^{(k)} = \left\| e_t^{(k)} \right\|_2$$

be the completion-error norm at time t. A temporal candidate edge $\{s, t\} \in E_k^{\mathrm{temp}}$ is retained when both endpoint errors are small:

$$\left\| e_s^{(k)} \right\|_2 \le \tau_e, \qquad \left\| e_t^{(k)} \right\|_2 \le \tau_e.$$

This rule connects nearby windows only when both windows are individually well predicted by the completion map.

4.2 Similar Errors

This rule ignores the absolute size of each individual error and only compares error-vector similarity. A temporal candidate edge is retained when

$$\left\| e_s^{(k)} - e_t^{(k)} \right\|_2 \le \tau_e.$$

The endpoint errors may be large; what matters is that the two error vectors are close to each other.

4.3 Lifted-Cloud Distance Graph

This rule uses the lifted points $p_t^{(k)} \in \mathbb{R}^{k_{\max}}$. For all temporal candidate edges, compute

$$d_{s,t}^{(k)} = \left\| p_s^{(k)} - p_t^{(k)} \right\|_2.$$

The edge $\{s, t\}$ is retained when

$$d_{s,t}^{(k)} \le \tau_d.$$

This is a local distance graph on the lifted test cloud, restricted to temporal candidate edges.

4.4 Window Covariance Similarity

Assume the AR process has finite innovation variance. The covariance rules are therefore not available for Cauchy innovations or for centered Pareto innovations with shape parameter at most 2.

Let

$$C_{s,t}^{(k)} = \mathrm{Cov}\left[Y_s^{(k)}, Y_t^{(k)} \right].$$

Using the stationary autocovariance function

$$\gamma(h) = \mathrm{Cov}(X_{u+h}, X_u),$$

the entries of this matrix are

$$C_{s,t}^{(k)} = \begin{pmatrix} \gamma(s-t) & \gamma(s-t-1) & \cdots & \gamma(s-t-k+1) \\ \gamma(s-t+1) & \gamma(s-t) & \cdots & \gamma(s-t-k+2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(s-t+k-1) & \gamma(s-t+k-2) & \cdots & \gamma(s-t) \end{pmatrix}.$$

The notebook computes γ from the AR coefficients by solving the stationary companion-matrix covariance equation, which is equivalent to the Yule-Walker equations.

The scalar covariance summary is

$$\rho_k(s,t) = \left\| C_{s,t}^{(k)} \right\|_F^2.$$

Because the graph convention retains edges with small weights, the notebook uses the reciprocal similarity weight

$$w_k^{\rho}(s,t) = \frac{1}{\rho_k(s,t) + \varepsilon}, \qquad \varepsilon = 10^{-12}.$$

Thus an edge is retained when

$$w_k^{\rho}(s,t) \le \tau_c.$$

Equivalently, this keeps temporally admissible pairs whose window covariance is large enough relative to the chosen threshold.

4.5 Lifted Covariance Similarity

The lifted point is

$$p_t^{(k)} = \begin{pmatrix} I_k \\ \widehat{A}_k \end{pmatrix} Y_t^{(k)}.$$

Therefore the cross-covariance matrix of two lifted points is

$$D_{s,t}^{(k)} = \mathrm{Cov}\left[p_t^{(k)}, p_s^{(k)}\right] = \begin{pmatrix} I_k \\ \widehat{A}_k \end{pmatrix} C_{s,t}^{(k)} \begin{pmatrix} I_k & \widehat{A}_k^{\top} \end{pmatrix}.$$

The scalar lifted-covariance summary is

$$\eta_k(s,t) = \left\| D_{s,t}^{(k)} \right\|_F^2.$$

The notebook again converts this similarity score into a threshold weight:

$$w_k^{\eta}(s,t) = \frac{1}{\eta_k(s,t) + \varepsilon}.$$

The edge $\{s,t\}$ is retained when

$$w_k^{\eta}(s,t) \le \tau_c.$$

This rule compares windows after the least-squares completion map has lifted them into the common ambient space $\mathbb{R}^{k_{\max}}$.

4.6 Full Temporal Graph

This rule retains every temporal candidate edge:

$$E_k = E_k^{\text{temp}}.$$

It is useful as a baseline because it shows what the flag complex looks like when the only restriction is temporal proximity.

5 Simple Functional Definitions

Let $K = L_k$ be one of the complexes. Let K_i be the set of i-simplices. In the notebook, dimensions $0, 1, 2$ correspond to vertices, edges, and triangles.

5.1 Face-Vector Entries

The face-vector entries are

$$f_i(K) = |K_i|.$$

The notebook keeps

$$f_0(K) = \text{number of vertices}, \qquad f_1(K) = \text{number of edges}, \qquad f_2(K) = \text{number of triangles}.$$

5.2 Betti Numbers

The Betti number $\beta_i(K)$ is the rank of the i-th homology group over $\mathbb{F}_2$:

$$\beta_i(K) = \operatorname{rank} H_i(K; \mathbb{F}_2).$$

The notebook keeps:

- $\beta_0(K)$, the number of connected components.
- $\beta_1(K)$, the number of independent one-dimensional cycles.

5.3 Point Complexity

Let $d = \dim K$, the largest simplex dimension present in K. For a vertex $x \in K_0$, the point complexity is

$$\operatorname{pc}(x) = d - \max\{\dim(\sigma) : \sigma \in K,\ x \in \sigma\}.$$

Thus a vertex has low point complexity when it belongs to a high-dimensional simplex, and high point complexity when it is isolated relative to the maximal dimension of the complex.

The notebook plots the total point complexity

$$\operatorname{PC}(K) = \sum_{x \in K_0} \operatorname{pc}(x).$$

Because the complexes are expanded through dimension two, the computed values use $d \leq 2$ in the current implementation.

5.4 Linear Combination of Two Functionals

In the interactive panel, the user chooses two basic functionals

$$F_1, F_2 \in \{f_0, f_1, f_2, \beta_0, \beta_1, \mathrm{PC}\}$$

and two scalar weights $a, b \in \mathbb{R}$. The plotted profile is

$$k \longmapsto aF_1(L_k) + bF_2(L_k).$$

For example, choosing $F_1 = \beta_0$, $F_2 = f_1$, $a = 2$, and $b = -1$ plots

$$2\beta_0(L_k) - f_1(L_k).$$

6 Scale-Similarity Matrix

The notebook also compares different embedding scales pairwise. It uses the same two functionals F_1, F_2 and scalar weights a, b chosen for the profile plot. Choose an operation

$$\oplus \in \{\cup, \cap\}.$$

The scale-similarity matrix is

$$S_{k\ell} = aF_1(L_k \oplus L_\ell) + bF_2(L_k \oplus L_\ell), \qquad 1 \le k, \ell \le k_{\max}.$$

The union $L_k \cup L_\ell$ contains all simplices appearing in either complex. The intersection $L_k \cap L_\ell$ contains exactly the simplices appearing in both complexes. Since all L_k are stored as simplicial complexes over the same test-window vertex set, both operations again produce simplicial complexes.

The matrix S is symmetric. The notebook computes an eigenvector

$$v = (v_1, \ldots, v_{k_{\max}})^\top$$

corresponding to the largest eigenvalue of S, and plots

$$k \longmapsto v_k.$$

The sign of an eigenvector is arbitrary, so the notebook flips the sign when necessary to make the sum of entries nonnegative.

7 Summary Table

Dropdown name	Mathematical meaning
`f0 (vertices)`	$f_0(K) = \lvert K_0 \rvert$
`f1 (edges)`	$f_1(K) = \lvert K_1 \rvert$
`f2 (triangles)`	$f_2(K) = \lvert K_2 \rvert$
`beta0`	$\beta_0(K)$
`beta1`	$\beta_1(K)$
`point complexity`	$\mathrm{PC}(K) = \sum_{x \in K_0} \mathrm{pc}(x)$